

SentinelX Project Report

AI-Powered Threat Intelligence and SOC Platform

1. Abstract

SentinelX is an integrated cybersecurity platform designed to support Security Operations Center (SOC) activities by combining threat intelligence, endpoint monitoring, detection, threat hunting, phishing investigation, honeypot monitoring, machine learning, artificial intelligence, incident response, and centralized visualization.

Modern organizations generate large volumes of security events from endpoints, networks, applications, and security tools. Security analysts often need to manually investigate suspicious IP addresses, domains, URLs, file hashes, alerts, and endpoint activities across multiple platforms. This fragmented workflow increases investigation time and makes it difficult to correlate security events with relevant threat intelligence.

SentinelX addresses this problem by providing a centralized architecture that collects and processes security information from multiple sources. Threat intelligence is aggregated from external providers such as AlienVault OTX, ThreatFox, AbuseIPDB, and VirusTotal. Indicators of Compromise (IOCs) are normalized, enriched, deduplicated, stored, and assigned risk scores.

The platform also integrates Wazuh and Sysmon to collect and analyze endpoint security events. A detection and correlation engine processes security events and threat intelligence to identify suspicious activities. Additional modules support proactive threat hunting, phishing email investigation, honeypot monitoring, machine learning-based threat prediction, and AI-assisted SOC investigations.

The overall objective of SentinelX is to demonstrate a unified and extensible foundation for intelligent SOC operations. The project combines multiple cybersecurity technologies into a single workflow that supports the process of collecting, enriching, detecting, investigating, and responding to security threats.

2. Introduction

Cybersecurity threats are continuously increasing in complexity and volume. Organizations are exposed to threats such as malware, phishing, unauthorized access, malicious IP addresses, suspicious network activity, and other advanced cyber attacks.

Security Operations Centers are responsible for monitoring security events, detecting threats, investigating suspicious activities, and coordinating incident response. However, security analysts frequently work with multiple independent tools such as Security Information and Event Management systems, endpoint monitoring solutions, threat intelligence platforms, phishing analysis tools, and incident response systems.

This fragmented approach creates several challenges. Analysts may need to manually search threat intelligence platforms, compare indicators, investigate endpoint logs, and correlate multiple alerts before determining whether an activity is malicious.

SentinelX is designed as an integrated cybersecurity platform that brings these capabilities together. The platform combines threat intelligence with security telemetry and provides a workflow for enrichment, risk scoring, detection, investigation, and response.

The project is structured using a modular architecture so that individual security capabilities can be developed and integrated independently.

The major components of SentinelX include:

- Threat Intelligence Aggregation
- IOC Normalization and Enrichment
- Risk Scoring
- Wazuh and Sysmon Integration
- Detection and Correlation Engine
- Threat Hunting
- Phishing Investigation
- Honeypot Monitoring
- Machine Learning
- AI SOC Agent
- SOC Dashboard
- Incident Response

3. Problem Statement

Modern organizations use multiple cybersecurity tools to monitor their infrastructure. Security data is often distributed across SIEM platforms, endpoint monitoring systems, threat intelligence sources, email security tools, and other security solutions.

This creates several problems:

- Security analysts must use multiple platforms.
- IOC investigation is often performed manually.
- Threat intelligence is not always automatically connected to security events.

- Security alerts may lack sufficient context.
- Correlation between different security events can be difficult.
- Incident investigation can be time-consuming.
- Analysts may receive a large number of alerts with different priorities.

There is a need for a centralized platform that can collect security information, enrich suspicious indicators with threat intelligence, prioritize risks, correlate events, support investigations, and assist analysts during incident response.

SentinelX is proposed to address this problem by providing an integrated SOC-oriented platform.

4. Objectives

The primary objectives of SentinelX are:

1. To develop a centralized cybersecurity and SOC platform.
2. To aggregate threat intelligence from multiple external sources.
3. To collect and normalize Indicators of Compromise.
4. To enrich IP addresses, domains, URLs, and file hashes.
5. To implement IOC risk scoring and severity classification.
6. To integrate Wazuh and Sysmon for endpoint security monitoring.
7. To develop a detection and correlation engine.
8. To support proactive threat hunting.
9. To investigate suspicious phishing emails.
10. To collect and analyze honeypot events.
11. To apply machine learning for threat prediction and classification.
12. To develop an AI-based SOC assistant.
13. To provide a centralized security dashboard.
14. To support the incident response lifecycle.
15. To create an extensible architecture for future cybersecurity modules.

5. Existing System

In a traditional security environment, different security operations are handled by separate tools.

For example:

- Threat intelligence is collected from external websites or APIs.
- Endpoint monitoring is performed using EDR or SIEM tools.
- Logs are analyzed through separate dashboards.
- Phishing emails are investigated manually.
- IOC reputation is checked using multiple services.

- Incident reports are created separately.

This approach has several limitations.

Limitations of Existing Systems

Fragmented Security Tools

Security teams may need to switch between multiple applications to investigate a single incident.

Manual IOC Enrichment

Analysts often manually search IP addresses, domains, URLs, and hashes across different threat intelligence platforms.

Limited Correlation

Security events from different systems may not automatically be connected.

Alert Overload

Large numbers of alerts can make it difficult to identify high-priority threats.

Slow Investigation

Manual data collection and enrichment increase the time required for analysis.

Limited Unified Visibility

Security information may be distributed across multiple dashboards.

These limitations demonstrate the need for an integrated security platform such as SentinelX.

6. Proposed System

SentinelX is proposed as a centralized platform for threat intelligence and SOC operations.

The system integrates multiple security capabilities into a unified workflow.

The proposed workflow is:

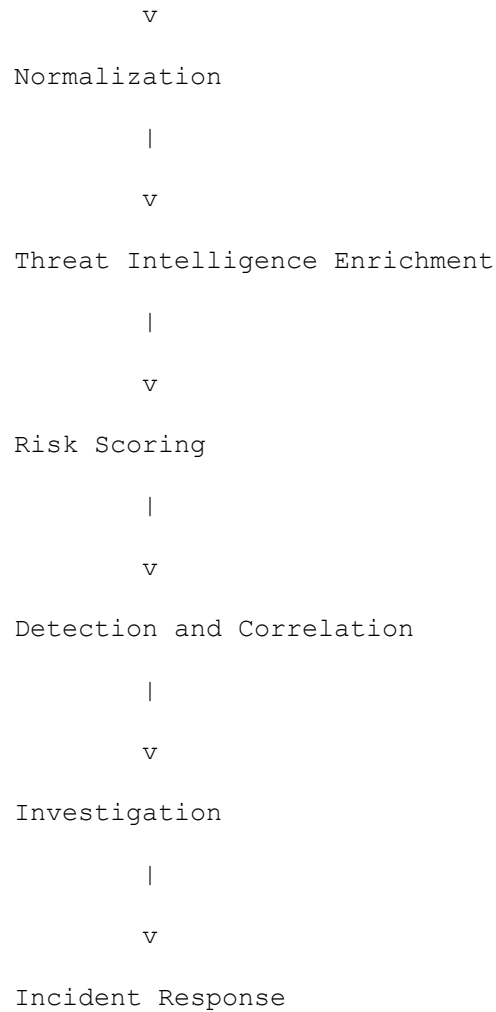
Security Data Sources

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Data Collection

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The system collects security information from multiple sources including:

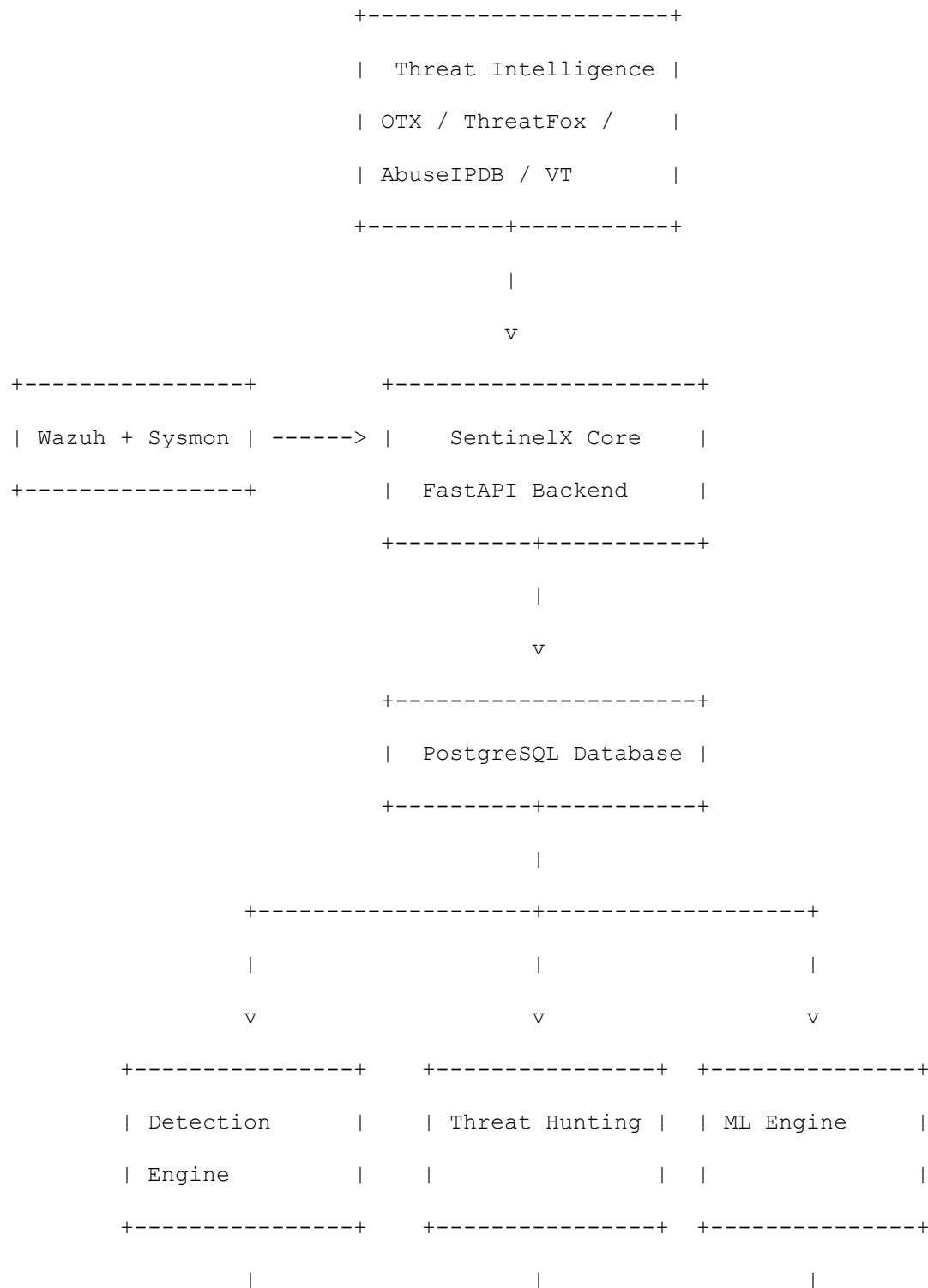
- Threat intelligence providers
- Wazuh
- Sysmon
- Phishing analysis
- Honeypot events
- Threat hunting queries

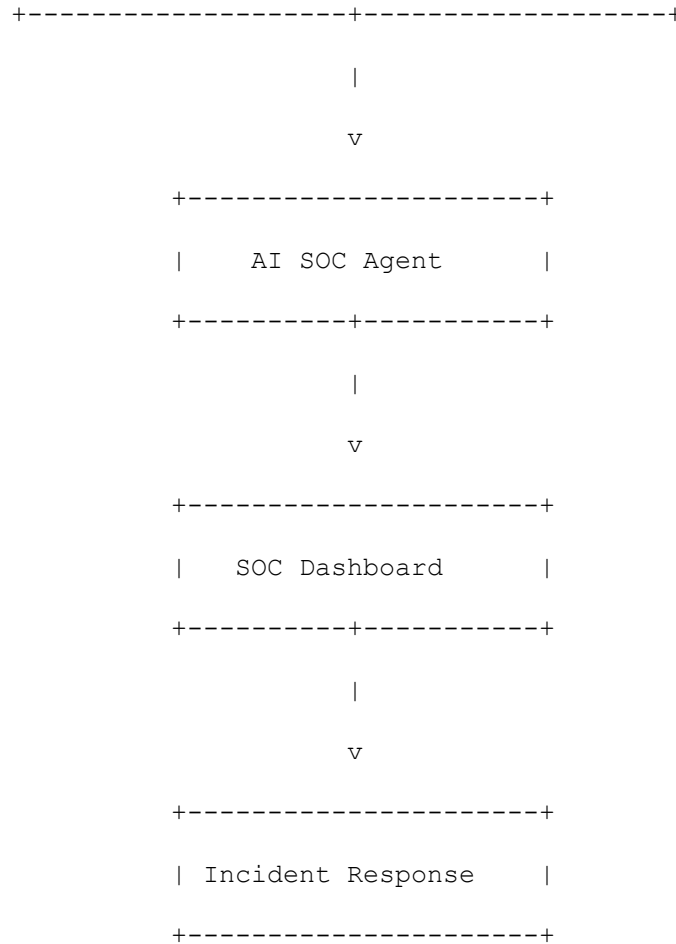
The collected information is processed and stored in a centralized database. Detection and analysis modules use this information to generate actionable security insights.

The proposed system also includes machine learning and AI-assisted investigation capabilities to improve analyst efficiency.

7. System Architecture

The SentinelX architecture is based on a modular cybersecurity workflow.





The architecture follows a layered approach:

1. Data Collection Layer
2. Processing and Normalization Layer
3. Threat Intelligence Layer
4. Detection and Analysis Layer
5. Investigation Layer
6. Visualization Layer
7. Incident Response Layer

8. Technologies Used

Python

Python is used for backend development, threat intelligence processing, automation, analysis, and machine learning.

FastAPI

FastAPI is used to develop REST APIs for SentinelX.

It provides:

- API endpoints
- Request validation
- Data models
- Backend services
- Integration between modules

PostgreSQL

PostgreSQL is used as the primary database for storing:

- IOCs
- Threat intelligence
- Analysis results
- Security events
- Alerts
- Incidents

React

React is used to build the SentinelX frontend interface and dashboard.

Vite

Vite is used as the frontend development and build tool.

Wazuh

Wazuh provides security monitoring and alerting capabilities.

Sysmon

Sysmon collects detailed endpoint telemetry from Windows systems.

Threat Intelligence APIs

SentinelX integrates threat intelligence from sources such as:

- AlienVault OTX
- ThreatFox
- AbuseIPDB

- VirusTotal

Machine Learning

Python-based machine learning components are used for threat prediction and classification.

Git and GitHub

Git is used for version control and GitHub is used for source code management.

9. Module Description

SentinelX consists of multiple modules.

Major Modules

1. Threat Intelligence Aggregator
2. IOC Normalization
3. IOC Enrichment
4. Risk Scoring
5. Wazuh and Sysmon Integration
6. Detection Engine
7. Threat Correlation
8. Threat Hunting
9. Phishing Investigation
10. Honeypot Integration
11. Machine Learning
12. AI SOC Agent
13. SOC Dashboard
14. Incident Response
15. Testing and Validation

Each module performs a specific cybersecurity function while integrating with the overall SentinelX architecture.

10. Threat Intelligence

The Threat Intelligence module collects security information from multiple external sources.

The purpose is to provide additional context for suspicious indicators.

Supported Threat Intelligence Sources

AlienVault OTX

Provides information about threat indicators and security pulses.

ThreatFox

Provides information related to malware-associated indicators.

AbuseIPDB

Provides reputation and abuse reports for IP addresses.

VirusTotal

Provides security analysis and reputation information for supported indicators.

Threat Intelligence Pipeline

Threat Source

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Data Collection

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Normalization

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Deduplication

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IOC Enrichment

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Risk Scoring

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Database Storage

The module helps transform raw threat intelligence into structured and actionable information.

11. Wazuh/Sysmon

SentinelX integrates Wazuh and Sysmon to provide endpoint security visibility.

Sysmon

Sysmon collects detailed Windows security events.

Examples include:

- Process creation
- Network connections
- File activity
- Registry modifications
- Process relationships

Wazuh

Wazuh collects and analyzes endpoint security information.

The general workflow is:

Windows Endpoint

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Sysmon

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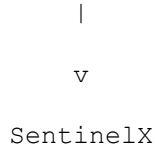
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Wazuh Agent

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Wazuh Manager



SentinelX uses relevant endpoint information to support enrichment, detection, and investigation.

12. Detection Engine

The Detection Engine processes security events and identifies suspicious activity.

The detection process includes:

1. Receive security events.
2. Normalize event information.
3. Extract relevant indicators.
4. Match indicators with threat intelligence.
5. Analyze severity and risk.
6. Correlate related events.
7. Generate a prioritized security result.

The workflow can be represented as:

Security Event



Normalization



IOC Matching



Threat Intelligence



Risk Analysis

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Correlation

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Security Alert

The objective is to provide analysts with actionable alerts instead of isolated raw events.

13. Threat Hunting

Threat hunting is a proactive security process used to search for hidden or undetected threats.

Instead of waiting only for alerts, analysts can investigate suspicious indicators and activities.

SentinelX supports threat hunting activities such as:

- Searching IP addresses
- Searching domains
- Investigating file hashes
- Investigating endpoint events
- Correlating security events
- Reviewing threat intelligence
- Identifying suspicious behavior

The threat hunting process follows:

Hypothesis

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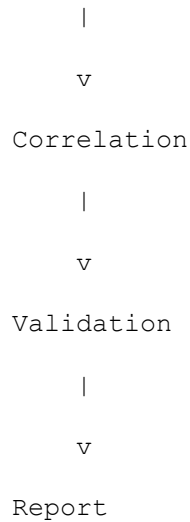
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Search

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Investigation



14. Phishing Investigation

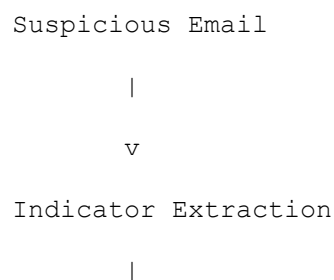
Phishing is one of the most common methods used by attackers to gain unauthorized access or distribute malicious content.

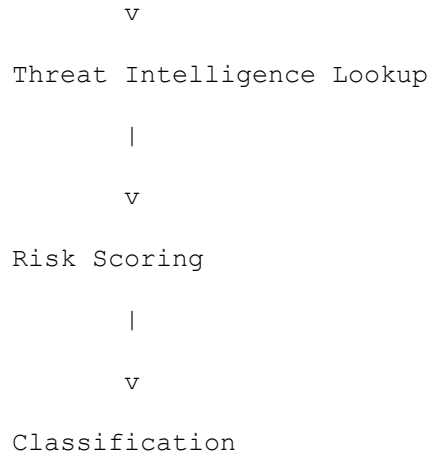
The Phishing Investigation module analyzes suspicious emails and extracts potential indicators.

The module can analyze:

- Sender information
- Email subject
- Email content
- URLs
- Domains
- IP addresses
- Attachments
- File hashes

The investigation workflow is:





The final result can help classify the email as:

- Benign
- Suspicious
- Malicious

15. Honeypot

A honeypot is a controlled decoy system designed to attract malicious activity.

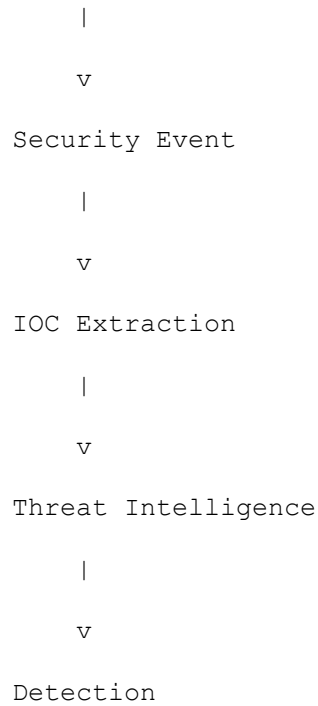
The Honeypot module can collect information about suspicious interactions.

Examples include:

- Source IP addresses
- Connection attempts
- Commands or actions
- Timestamps
- Attack patterns
- Extracted indicators

The workflow is:





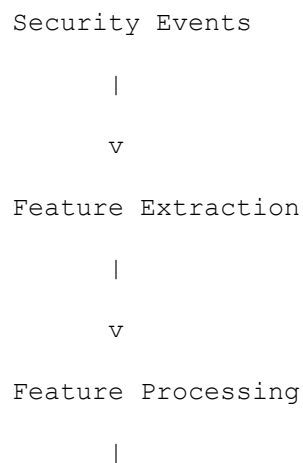
The collected information can be used to improve threat detection and investigation.

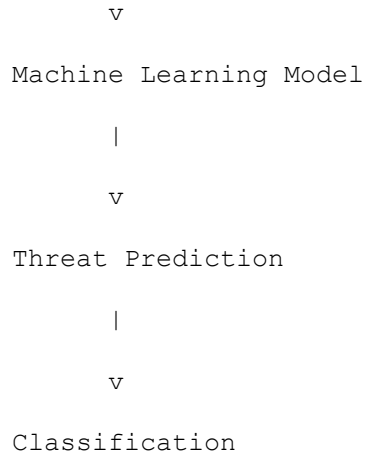
16. Machine Learning

Machine learning can provide an additional layer of analysis for cybersecurity events.

The ML module processes security features and attempts to estimate the likelihood that an event is suspicious or malicious.

The general pipeline is:





Potential applications include:

- Suspicious event classification
- Risk prioritization
- Anomaly identification
- Threat prediction

Machine learning results should be combined with other evidence such as:

- Detection rules
- Threat intelligence
- Event correlation
- Analyst validation

This reduces the risk of relying entirely on automated predictions.

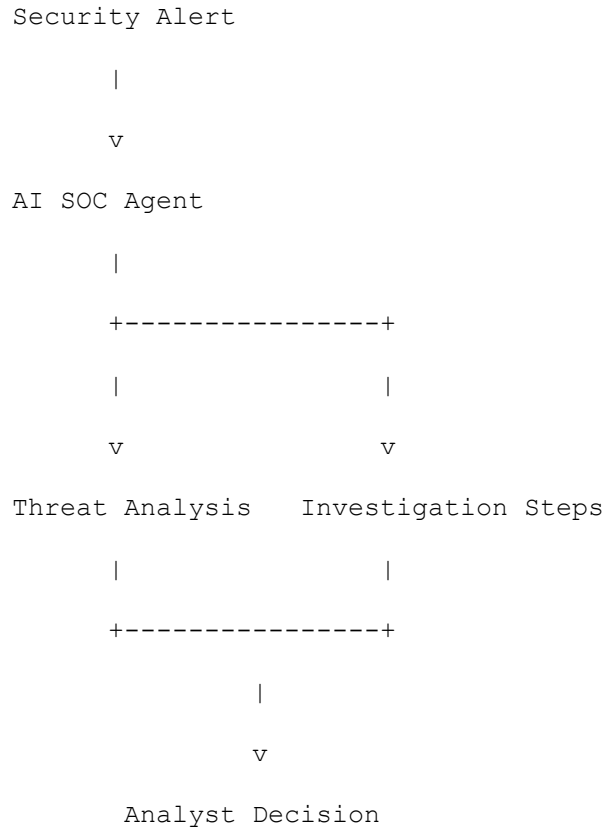
17. AI SOC Agent

The AI SOC Agent is designed to assist security analysts during investigations.

The agent can support tasks such as:

- Alert summarization
- IOC investigation
- Threat intelligence explanation
- Investigation guidance
- Response recommendations
- Incident summarization

Example workflow:



The AI SOC Agent is intended to support human analysts rather than independently perform uncontrolled security actions.

18. Dashboard

The SentinelX dashboard provides centralized visibility into security operations.

The dashboard can display:

- System status
- Security alerts
- IOC information
- Threat intelligence results
- Severity levels
- Detection results
- Threat hunting information
- Investigation data
- Incident information

The objective is to provide a single interface where analysts can monitor and investigate security activity.

The dashboard connects the backend APIs with the SOC analyst workflow.

19. Incident Response

The Incident Response module supports the process of managing security incidents.

The incident lifecycle includes:

Detection

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Triage

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Investigation

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Containment

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Eradication

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Recovery

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Lessons Learned

SentinelX can support incident response by providing:

- Alert prioritization
- IOC enrichment
- Evidence collection
- Investigation context
- Incident tracking
- Response recommendations
- Incident summaries and reports

20. Testing

Testing was performed to validate the functionality of SentinelX modules and integrations.

Testing Areas

Backend Testing

FastAPI endpoints were tested to verify API functionality.

Database Testing

PostgreSQL connectivity and data storage were tested.

Threat Intelligence Testing

Threat intelligence collection and aggregation workflows were tested.

IOC Storage Testing

IOC information was tested for successful storage and retrieval.

Normalization Testing

Different indicators were tested to ensure consistent internal formatting.

Risk Scoring Testing

IOC and analysis scoring functionality was validated.

Wazuh Integration Testing

Wazuh-related security information was tested as part of the integration workflow.

Sysmon Testing

Endpoint events generated by Sysmon were used to validate monitoring.

Detection Testing

Detection and correlation workflows were tested using available security events and indicators.

End-to-End Testing

The complete workflow was tested from data ingestion through analysis and storage.

21. Results

The SentinelX project successfully demonstrates the core components of an integrated SOC platform.

The following results were achieved:

- FastAPI backend developed and tested.
- PostgreSQL database integrated successfully.
- Threat intelligence aggregation implemented.
- IOC normalization implemented.
- IOC enrichment implemented.
- Threat intelligence information stored successfully.
- Risk scoring and severity classification implemented.
- Wazuh and Sysmon integration workflow established.
- Detection and analysis workflows implemented.
- Threat hunting functionality developed.
- Phishing investigation architecture implemented.
- Honeypot integration module designed.
- Machine learning threat analysis functionality developed.
- AI SOC Agent functionality designed and integrated.
- SOC dashboard developed.
- Incident response workflow integrated into the platform.

The project demonstrates the feasibility of combining multiple cybersecurity capabilities into a unified platform.

22. Limitations

Although SentinelX provides an integrated cybersecurity platform, there are several limitations.

Limited Threat Intelligence Coverage

The platform depends on the availability and quality of external threat intelligence sources.

API Limitations

Some threat intelligence providers may require API keys, rate limits, or paid subscriptions.

Machine Learning Data

The accuracy of machine learning depends on the quality and size of available training data.

False Positives

Detection rules and ML models may generate false positive alerts.

Lab Environment

The current implementation is primarily developed and tested in a controlled laboratory environment.

Scalability

Large enterprise environments may require distributed processing and scalable infrastructure.

Automation Risks

Automated response actions must be carefully controlled to avoid unintended effects.

23. Future Scope

SentinelX can be extended in several directions.

Advanced Machine Learning

Use larger datasets and advanced models for improved anomaly detection and threat classification.

Additional Threat Intelligence

Integrate additional commercial and open-source threat intelligence providers.

MITRE ATT&CK Mapping

Map security events and detections to MITRE ATT&CK techniques.

SOAR Integration

Implement Security Orchestration, Automation and Response capabilities.

Automated Response

Develop controlled response playbooks for selected security events.

Examples may include:

- Blocking malicious IP addresses
- Isolating endpoints
- Disabling compromised accounts
- Blocking malicious domains

Cloud Deployment

Deploy SentinelX using cloud infrastructure.

Containerization

Use Docker and Kubernetes for scalable deployment.

Advanced Visualization

Add real-time attack visualization and security analytics.

Multi-Tenant Architecture

Support multiple organizations or customers in a managed SOC environment.

Continuous Attack Simulation

Use controlled attack simulations to continuously validate detections and response workflows.

24. Conclusion

SentinelX is an integrated cybersecurity platform designed to demonstrate a unified approach to Security Operations Center activities.

The project combines threat intelligence, endpoint monitoring, detection, threat hunting, phishing investigation, honeypot monitoring, machine learning, artificial intelligence, dashboard visualization, and incident response.

The Threat Intelligence module aggregates information from multiple sources and enriches Indicators of Compromise. Wazuh and Sysmon provide endpoint security visibility. The Detection Engine processes and correlates security information to identify suspicious activity. Threat hunting enables proactive investigation, while phishing and honeypot modules provide additional security visibility.

Machine learning adds an analytical layer for threat classification and prioritization. The AI SOC Agent assists analysts by summarizing security information and suggesting investigation steps. The centralized dashboard provides a unified interface for monitoring and investigation.

Overall, SentinelX demonstrates how multiple cybersecurity technologies can be integrated into a modular SOC platform.

The project provides a foundation that can be further expanded with advanced machine learning, automated response, MITRE ATT&CK mapping, cloud deployment, SOAR integration, and enterprise-scale architecture.

SentinelX demonstrates the concept of an intelligent, threat intelligence-driven SOC platform capable of supporting modern cybersecurity monitoring, detection, investigation, and incident response.